

Deepak Mallampati

FORWARD DEPLOYED ENGINEER

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

AI/ML engineer with 4+ years building and deploying production Generative AI for healthcare and regulated enterprises. Ships client-facing RAG and agentic LLM systems end to end - from data and retrieval to FastAPI services in production - cutting hallucinations by 40% at 92% precision across millions of regulatory documents. Strong on compliant deployment (HIPAA, PII redaction, audit logging) and 0-1 delivery, working directly with stakeholders to integrate tailored AI solutions and sustain 99.9% uptime.

SKILLS

Forward Deployed / Client-Facing Delivery • Healthcare AI & RAG • Agentic LLM Workflows • FastAPI & Microservices • EHR / Regulated-Data Integration • 0-1 Product Shipping • HIPAA & Compliance

Generative AI & LLMs: GPT-4o, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Prompt Engineering | Fine-tuning (LoRA, QLoRA, PEFT) | Content Safety

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | Semantic Caching

ML & NLP: PyTorch, TensorFlow, Keras | NER, Classification, Summarization | RAGAS, BERTScore | Drift Detection | A/B Testing

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda), Azure (OpenAI, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B | Monitoring & Observability

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | PostgreSQL, MongoDB, Redis | Git, Jira | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Engineered high-performance **RAG** pipelines over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy), enabling compliant analytics and model development with zero PII exposure.
- Built fault-tolerant automation infrastructure with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Established an **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency/accuracy dashboards to catch regressions before release.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Built **12 APIs and microservices** (FastAPI) powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by **40%**.
- Developed AI agents and backend services for Vanguard's internal AI platform across **25+ AI agents** and workflows, integrating with production data pipelines and observability tooling.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing model-selection decisions.
- Engineered content controls for policy-compliant AI responses, reducing unsafe outputs by **42%**; delivered scalable LLM apps supporting **100,000+ requests daily** across 20+ teams.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a **privacy-preserving ML pipeline** on a clinical hospital-readmission dataset (k-anonymity, l-diversity, Laplace differential privacy across four ϵ levels).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%** and eliminated unique-record exposure while retaining **99% of baseline predictive performance**.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Developed supervised regression and classification models for equipment-failure prediction, anomaly detection, and capacity forecasting on operational pipeline-storage data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on financial datasets, achieving **89% F1-score** on custom NER tasks.

PROJECTS

Privacy-Preserving Clinical ML Pipeline

A framework that lets researchers analyze sensitive patient data without exposing identities, combining k-anonymity, l-diversity, and differential privacy - quantifying exactly how much privacy each technique buys for how little accuracy cost.

career-ops

An autonomous, AI-driven job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - compressing hours of repetitive work into an automated agentic workflow.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across 29+ sites, turning a multi-step manual chore into a single inline action.

RESEARCH

Authored a peer-review-ready **IEEE-format conference paper** (LaTeX) on a privacy-utility scoring framework for clinical anonymization, benchmarking six configurations and quantifying differential-privacy utility cost across ϵ levels.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70 / 4.0